Test 4 – Grade 8

Name: _____
Grade / Class: 8

Section A: English Language & Literature

1. Reading Comprehension
Read the passage and answer the questions:

"During the winter break, Ahmed volunteered at a local library. He organized books, assisted younger students in reading activities, and created posters promoting literacy in the community. By the end of the month, Ahmed felt satisfied with his contribution and noticed the library became livelier with more young visitors."

a) Where did Ahmed volunteer during the winter break?
b) List two activities Ahmed performed at the library.
c) How did Ahmed feel after volunteering?
d) Summarize the passage in four sentences using your own words.

2. Grammar & Vocabulary
a) Combine the sentences using an appropriate conjunction:
"The storm was severe. We managed to reach home safely."

b) Identify the nouns in the sentence:
"The diligent students completed their homework before the teacher arrived."

c) Correct the error in this sentence:
"Each of the girls have completed the assignment."

d) Convert into indirect speech:
"He said, 'I am going to participate in the science fair.'"

3. Writing
Write a short essay (120–150 words) on the topic: "The Impact of Social Media on Teenagers". Include positive and negative effects, and your personal opinion.

Section B: Mathematics

1. Algebra
a) Solve for x:
$7x - 14 = 28$

b) Factorize:
$x^2 - 11x + 30$

c) Simplify:
$8p^2q - 5pq^2 + 3p^2q$

2. Geometry & Mensuration
a) The base of a triangle is 18 cm and the height is 10 cm. Find the area.
b) A cylinder has a radius of 6 cm and height 12 cm. Find the volume.

3. Ratio, Proportion & Percentage
a) A laptop costs $800. It is sold at a 20% discount. Find the selling price.
b) If 12 notebooks cost $36, find the cost of 25 notebooks.

4. Data Handling
Marks of 10 students in a science test: 10, 15, 12, 18, 20, 15, 16, 14, 18, 17.
a) Find the mean, median, and mode.
b) Represent the data using a simple frequency table.

Section C: Science

1. Biology
a) Explain the functions of the human liver.
b) Define respiration and distinguish between aerobic and anaerobic respiration.
c) List two differences between reptiles and amphibians.

2. Physics
a) A cyclist accelerates from 2 m/s to 10 m/s in 4 seconds. Calculate the acceleration.
b) Define potential energy and give one example.
c) Explain the difference between reflection and refraction of light.

3. Chemistry
a) Identify elements, compounds, or mixtures:

- **Carbon dioxide**
- **Iron**
- **Saltwater**

b) Write a balanced chemical equation for the reaction between zinc and hydrochloric acid.

4. Environmental Science
a) List two sources of water pollution and their effects.
b) Suggest three ways to reduce energy consumption at home.

Section D: Social Studies

1. History
a) Name two major revolutions of the 19th century and their outcomes.
b) Explain the importance of the Industrial Revolution.
c) Identify the countries forming the Axis Powers during World War II.

2. Geography
a) Define longitude and explain how it influences time zones.
b) Name the largest rainforest in the world and its location.
c) Discuss two environmental effects of urbanization.

3. Civics / Political Science
a) Differentiate between democracy and monarchy.
b) State two civic responsibilities of students in a school setting.

4. Economics
a) Explain the law of demand with an example.
b) Distinguish between gross income and net income.

Section E: ICT / Computer Studies

1. Computer Basics
a) Define the internet and name one advantage.
b) Name one input device and one storage device.

2. Applications
a) Which spreadsheet function calculates the average of a dataset?
b) Give two ways presentation software is useful for teachers.

3. Programming Basics
a) Define a function in programming with an example.
b) Write Python code to calculate the sum of numbers from 1 to 10.

4. Cyber Awareness
a) Explain phishing and one way to prevent it.
b) Mention two methods to secure passwords online.

Section F: Islamiyat

1. Quran & Hadith
a) Name two teachings from the Quran about kindness.
b) Narrate a Hadith encouraging honesty.

2. Aqidah / Belief
a) List the six pillars of Iman (faith).
b) Explain why Tawheed is fundamental in Islam.

3. Fiqh / Worship
a) Explain the importance of Zakah.
b) Describe the significance of fasting in Ramadan.

4. Islamic Ethics & Morality
a) How does Islam encourage fairness in trade?
b) Explain the importance of protecting natural resources according to Islam.

Section G: Arts & Music

1. Visual Arts
a) Explain symmetry in art.
b) Name two secondary colors.

2. Music
a) Define rhythm and tempo.
b) Give an example of a percussion instrument.

Section H: Health & Physical Education

1. Physical Fitness
a) Name two exercises to strengthen core muscles.
b) List one competitive sport and one recreational activity.

2. Health & Hygiene
a) Why is hydration important for teenagers?
b) Mention two habits to prevent infectious diseases.

3. Safety & Environment
a) State one rule for safe playground activities.